

RAHUL MITRA

SOFTWARE ENGINEER PROFILE

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PROFILE SUMMARY

Early-career graduate software engineer with NUS ISE, Computer Science, and Mathematics training, building production-minded systems across Python, React, TypeScript, cloud deployment, AI-assisted development, and open-source tooling.

EDUCATION

National University of Singapore

Aug 2023 - Jul 2027

- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics
- Relevant coursework: Software Engineering; 3D Computer Vision; Artificial Intelligence; Data Structures and Algorithms; Simulation; Statistics; Operations Research
- Certificate of Outstanding Performance in 3D Computer Vision (CS4277) - top student in a class of 24, NUS School of Computing.

Anderson Serangoon Junior College

Feb 2019 - Dec 2020

- Singapore-Cambridge GCE A-Level

WORK AND INTERNSHIP EXPERIENCE

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories Singapore Pte Ltd Jun 2026 - Present

- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.

Data Analytics and Project/Product Management Intern, Abbott Laboratories Singapore Pte Ltd Jan 2026 - Jun 2026

- Built a SimPy discrete-event digital twin and CP-SAT scheduling optimizer; engineered a non-destructive 15-stage data pipeline that handled human- and machine-generated errors and processed five years of unseen raw data without errors.
- Productionized and operated a team-built APC simulator for live internal manufacturing and engineer-training users, resolving runtime issues and packaging it as a Docker container for Azure App Service; delivered regional AI upskilling.

Software Engineer (Citizen Developer), People's Association

Sep 2025 - Present

- Built an end-to-end digital garden-plot balloting platform (admin and public-facing frontends, backend, GIS mapping, and deployment), now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks.
- Led deployment planning, multi-repo coordination, and technical project management across product, engineering, and operations stakeholders; awarded the S\$20,000 Sparks Community Innovation Fund.

Base Support Assistant, Republic of Singapore Navy

Jan 2022 - Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.
- Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.

SELECTED PROJECTS AND COMPETITIONS

AsyncDDGS - Open-Source Search Client

2024 - Present

- Maintained an asyncio-first, aiohttp-based DuckDuckGo search client on PyPI with pytest CI/CD and adoption across AI tooling, Discord bots, and search-enabled projects.

Interactive Portfolio Infrastructure

2026

- Built a React, TypeScript, Three.js, Matter.js, and AI-assisted portfolio with profile switching, searchable field notes, analytics, and a 3D project world.

OnTheSpectrum - Codex and Blender MCP 3D Pipeline

May 2026

- Built an innovative local-first Blender-to-Three.js asset pipeline using Codex, Blender MCP, generated GLB assets, metadata, previews, playable world QA, and Remotion video creation workflows.

SmartExam - GovTech x NTUPC Product Hackathon

2025

- Built a multi-agent autonomous exam generation system with RAG, Next.js, source ingestion, question generation, and education workflow automation.

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: Python; TypeScript; React; Next.js; Three.js; FastAPI; aiohttp/asyncio; Docker; CI/CD; Azure; 3D CV: projective & epipolar geometry, absolute pose, SfM, bundle adjustment, multi-view stereo.

- NVIDIA Disaster Risk Monitoring Using Satellite Imagery - geospatial AI, U-Net, satellite data, GPU workflows.
- Packt Deep Reinforcement Learning Hands-On - DQN, Double DQN, DDPG, PPO, A2C, Python implementation.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.

Astronomy Head and Robotics Member, STEM Inc. (ASRJG)

2019 - 2020

- Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.